

Operating Systems — Labs

Problem Set 1: Grade Lab Utilities

```
aadeshkp@aadeshkp:~/Misc/xv6-labs-2025$ ./grade-lab-util
make: 'kernel/kernel' is up to date.
== Test sleep, no arguments == sleep, no arguments: OK (0.8s)
    (Old xv6.out.sleep_no_args failure log removed)
== Test sleep, returns == sleep, returns: OK (0.9s)
== Test sleep, makes syscall == sleep, makes syscall: OK (0.9s)
    (Old xv6.out.sleep failure log removed)
== Test sixfive_test == sixfive_test: OK (1.1s)
    (Old xv6.out.sixfive_test failure log removed)
== Test sixfive_readme == sixfive_readme: OK (1.5s)
    (Old xv6.out.sixfive_readme failure log removed)
== Test sixfive_all == sixfive_all: OK (1.2s)
== Test memdump, examples == memdump, examples: OK (0.5s)
== Test memdump, format ii, S, p == memdump, format ii, S, p: OK (0.9s)
== Test find, in current directory == find, in current directory: OK (0.9s)
== Test find, in sub-directory == find, in sub-directory: OK (1.1s)
== Test find, recursive == find, recursive: OK (1.2s)
== Test exec == exec: OK (0.7s)
== Test exec, multiple args == exec, multiple args: OK (1.0s)
== Test exec, recursive find == exec, recursive find: OK (1.3s)
== Test time ==
time: FAIL
    Cannot read time.txt
Score: 130/131
```

Output:

1. sleep.c

```
aadeshkp@aadeshkp:~/Misc/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo u
ser/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir use
r/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie
user/_logstress user/_forphan user/_dorphan user/_sleep user/_sixfive user/_memd
ump user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1
953 total 2000
balloc: first 1090 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sleep 100
$ █
```

2. sixfive.c

```
aadeshkp@aadeshkp: ~/Misc/xv6-labs-2025
ser/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir use
r/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie
user/_logstress user/_forphan user/_dorphan user/_sleep user/_sixfive user/_memd
ump user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1
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qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sleep 100
$ sixfive sixfive.txt
5
100
18
6
```

3. memdump.c

```
aadeshekp@aadeshekp: ~/Misc/xv6-labs-2025
18
6
$ memdump
Example 1:
61810
2025
Example 2:
a string
Example 3:
another
Example 4:
B38
1819438967
100
z
xyzzz
Example 5:
hello
w
o
r
l
d
$

another
Example 4:
B38
1819438967
100
z
xyzzz
Example 5:
hello
w
o
r
l
d
$ echo deadc0de | memdump hhcccc
25956
25697
c
0
d
e
$ echo deadc0de | memdump p
64616564
$
```

4. find.c

```
aadeshkp@aadeshkp: ~/Misc/xv6-labs-2025
w
o
r
l
d
$ echo deadc0de | memdump hhcccc
25956
25697
c
0
d
e
$ echo deadc0de | memdump p
64616564
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$
```

5. find.c (with exec extension)

```
aadeshkp@aadeshkp: ~/Misc/xv6-labs-2025
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo u
ser/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir use
r/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie
user/_logstress user/_forphan user/_dorphan user/_sleep user/_sixfive user/_memd
ump user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1
953 total 2000
ballocc: first 1090 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sh < findtest.sh
$ $ $ $ $ $ hello
hello
hello
$ $
```

